

Phase 0: Mathematical Reboot

A diagnostic and refresher workbook for the Chebotarev Density
study course

Proof fluency, algebra, Galois theory, real and complex analysis

Purpose. Establish whether your old graduate-level mathematical machinery is accessible enough to begin the commutative-algebra and algebraic-number-theory phases. This is not a prerequisite marathon. It is a focused reboot.

Complete exercises first without solutions; then use the full solutions for diagnosis and repair.

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How to use this workbook

Phase 0 has one job: restore enough mathematical fluency that later material can be learned for its mathematical content rather than being blocked by forgotten notation, elementary lemmas, or proof habits.

This workbook is designed for roughly **10–20 focused hours**. It is deliberately selective. You do not need to become an expert again in every topic represented here before moving on.

Method

Recommended sequence

1. Take the Baseline Diagnostic in Chapter 1 closed-book. Allow 2–3 hours and mark uncertain answers explicitly.
2. Read Chapters 2–6 selectively. Spend more time where the diagnostic exposed weakness.
3. Solve the practice problems in each chapter before consulting Part III.
4. Take the Final Readiness Test in Chapter 8 closed-book.
5. Use the scoring rubric in Chapter 9 to decide whether to begin Phase 1 or spend several more days repairing specific gaps.

Checkpoint

The advancement criterion is functional, not encyclopedic. Move on when you can read graduate mathematics comfortably; state the central definitions accurately; reconstruct representative arguments; and solve standard examples without constant reference to notes. Forgetting a theorem name or a technical trick is not a reason to remain in Phase 0.

Chapter 1

Baseline diagnostic: closed book

Do this chapter before reading the refreshers. The purpose is to distinguish *rust* from genuine gaps. Rust usually looks like “I know what should happen but cannot quite execute.” A genuine gap looks like “I do not know what object or theorem is relevant.” The latter deserves more review.

For each problem, write one of **C** (confident), **U** (uncertain), or **N** (no workable approach) before beginning. This self-rating is useful later: mathematical recovery often first appears as better calibration.

1.1 Algebra and Galois theory

D1. Quotient field criterion

Prove that $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime.

D2. Chinese remainder theorem in action

Find the unique residue class $x \pmod{105}$ satisfying

$$x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{5}, \quad x \equiv 4 \pmod{7}.$$

Then state the ring-theoretic Chinese remainder theorem for pairwise comaximal ideals.

D3. Homomorphism and quotient

Let $\varphi : G \rightarrow H$ be a group homomorphism. Prove that $\ker \varphi \trianglelefteq G$ and construct an explicit isomorphism

$$G/\ker \varphi \cong \text{im } \varphi.$$

D4. Group action

Let S_3 act on $\{1, 2, 3\}$ in the natural way. Determine the orbit and stabilizer of 1, and verify the orbit–stabilizer formula. Then describe the conjugacy classes of S_3 .

D5. Sylow reasoning

Let G be a group of order 21. Show that the Sylow 7-subgroup is normal. What can you say about the number of Sylow 3-subgroups?

D6. Prime and maximal ideals

Let R be a commutative ring with identity and $I \subsetneq R$ an ideal. Prove:

- (a) I is prime if and only if R/I is an integral domain;
- (b) I is maximal if and only if R/I is a field.

D7. Minimal polynomial

Find the minimal polynomial over $\mathbb{Q}$ of $\alpha = \sqrt{2} + \sqrt{3}$. Determine $[\mathbb{Q}(\alpha) : \mathbb{Q}]$.

D8. Finite fields and Frobenius

Let $q = p^r$. Prove that the map

$$F : \mathbb{F}_{q^n} \rightarrow \mathbb{F}_{q^n}, \quad F(x) = x^q,$$

is an automorphism fixing $\mathbb{F}_q$. Show that $F^n = \text{id}$.

D9. Galois correspondence

Let $L/\mathbb{Q}$ be the splitting field of $x^3 - 2$. Determine $[L : \mathbb{Q}]$ and $\text{Gal}(L/\mathbb{Q})$ up to isomorphism. Identify the fixed field of the subgroup generated by complex conjugation.

1.2 Analysis

D10. Series and comparison

Determine whether each series converges absolutely, conditionally, or diverges:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}, \quad \sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}, \quad \sum_{n=1}^{\infty} \frac{n!}{n^n}.$$

Justify your answers.

D11. Pointwise versus uniform convergence

On $[0, 1]$, let $f_n(x) = x^n$. Find the pointwise limit. Does f_n converge uniformly? Prove your answer.

D12. Uniform convergence and integration

Suppose $f_n : [a, b] \rightarrow \mathbb{R}$ are continuous and $f_n \rightarrow f$ uniformly. Prove that

$$\int_a^b f_n(x) dx \longrightarrow \int_a^b f(x) dx.$$

D13. Radius of convergence

Find the radius of convergence of

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n} z^n.$$

What can you say about convergence on the boundary?

D14. Cauchy integral formula

Evaluate

$$\int_{|z|=2} \frac{e^z}{z(z-1)} dz,$$

where the circle is traversed counterclockwise.

D15. Residues

Compute

$$\operatorname{Res}_{z=0} \frac{e^z}{z^3} \quad \text{and} \quad \operatorname{Res}_{z=i} \frac{1}{(z^2+1)^2}.$$

1.3 Proof fluency

D16. Infinitely many primes

Give a complete proof that there are infinitely many prime numbers. Then identify exactly where unique factorization is being used, if at all.

D17. Root bound for polynomials

Let F be a field and $0 \neq f \in F[x]$ have degree d . Prove that f has at most d roots in F .

D18. Compactness-style proof

Let (x_n) be a bounded sequence in $\mathbb{R}$. State the Bolzano–Weierstrass theorem precisely and explain how it differs logically from the assertion that every bounded sequence converges.

Chapter 2

Proof language and proof reconstruction

The main obstacle after a long break is often not missing knowledge but loss of *proof compression*: statements that once immediately suggested a proof pattern now require conscious unpacking. This chapter restores those patterns.

2.1 Quantifiers are part of the mathematics

A statement such as

$$\forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N : |x_n - L| < \varepsilon$$

is not merely formal decoration. Its order tells you what your proof is allowed to depend on. The index N may depend on ε , but not on n .

Negating quantified statements should become automatic:

$$\neg(\forall x \exists y P(x, y)) \iff \exists x \forall y \neg P(x, y).$$

For example, $f_n \rightarrow f$ uniformly on E means

$$\forall \varepsilon > 0 \exists N \forall n \geq N \forall x \in E : |f_n(x) - f(x)| < \varepsilon.$$

Failure of uniform convergence means that there exists an $\varepsilon_0 > 0$ such that for every N there exist $n \geq N$ and $x \in E$ with

$$|f_n(x) - f(x)| \geq \varepsilon_0.$$

That negation is often the easiest way to prove non-uniformity.

2.2 Four proof patterns to reactivate

2.2.1 Direct proof

To prove $P \Rightarrow Q$, assume P , unfold the definitions, and derive Q . Quotient-ring characterizations of prime and maximal ideals are archetypal examples.

2.2.2 Contrapositive

Instead of $P \Rightarrow Q$, prove $\neg Q \Rightarrow \neg P$. This is often useful when $\neg Q$ produces an explicit witness.

2.2.3 Contradiction

Assume the desired conclusion is false and derive an impossibility. Euclid's proof of infinitely many primes is the classical example.

2.2.4 Induction

Use induction only when the problem has an actual recursive or ordered structure. Do not reach for it merely because an integer n appears.

Intuition

A proof is usually easiest when you ask, "What would count as a witness?" For existential statements, build the witness. For universal statements, choose an arbitrary object. For equality of sets, prove two inclusions. For equality of functions, evaluate at an arbitrary input. For equality of subgroups or fields, prove containment in both directions or compare finite indices/degrees.

2.3 Proof reconstruction rather than rereading

When reviewing a theorem, use this cycle:

1. Read the statement until every hypothesis has a purpose you can at least guess.
2. Close the source and write a 3–6 line proof skeleton.
3. Reopen the source only to repair missing steps.
4. The next day, reconstruct the proof from the skeleton alone.

The goal is not verbatim memory. It is to retain the *architecture*: what object is constructed, what theorem is invoked, and why the conclusion follows.

P1. Negating uniform convergence

Write the logical negation of the statement " $f_n \rightarrow f$ uniformly on E ". Then use that negation to prove that $f_n(x) = x^n$ does not converge uniformly on $[0, 1]$.

P2. Equality by two inclusions

For a group homomorphism $\varphi : G \rightarrow H$ and subgroups $A, B \leq G$, prove

$$\varphi(A \cap B) \subseteq \varphi(A) \cap \varphi(B).$$

Give an explicit example where the inclusion is strict.

P3. Hypothesis audit

Consider the theorem: "A continuous real-valued function on a compact set attains its maximum." For each hypothesis (continuity, compactness), give an example showing the conclusion can fail when that hypothesis is removed while the other is retained.

Chapter 3

Algebra reboot I: groups, actions and ideals

3.1 Groups and homomorphisms

Definition 3.1. A *group* is a set G with an associative binary operation, an identity element, and inverses. A subgroup $H \leq G$ is *normal*, written $H \trianglelefteq G$, if $gHg^{-1} = H$ for every $g \in G$.

Normality is exactly what is required to multiply cosets consistently:

$$(gH)(kH) = (gk)H.$$

Kernels are normal because a homomorphism cannot distinguish elements in the same kernel coset.

Theorem 3.2 (First isomorphism theorem). *For a homomorphism $\varphi : G \rightarrow H$, the map*

$$G / \ker \varphi \longrightarrow \operatorname{im} \varphi, \quad g \ker \varphi \longmapsto \varphi(g)$$

is a well-defined isomorphism.

Intuition

A homomorphism throws away exactly the information contained in its kernel. Quotienting by the kernel identifies precisely those elements the map already treats as equal.

3.2 Group actions

An action of G on a set X is a homomorphism $G \rightarrow S_X$, where S_X is the permutation group of X . For $x \in X$, define

$$Gx = \{gx : g \in G\}, \quad G_x = \{g \in G : gx = x\}.$$

Theorem 3.3 (Orbit–stabilizer). *If G is finite, then*

$$|G| = |Gx| |G_x|.$$

Conjugation is the action

$$g \cdot x = gxg^{-1}.$$

Its orbits are conjugacy classes. This matters later because Frobenius elements attached to different primes above the same base prime are conjugate.

3.2.1 Sylow theorems: what you need to remember

If $|G| = p^a m$ with $p \nmid m$, then:

- Sylow p -subgroups have order p^a ;
- they exist;
- they are all conjugate;
- their number n_p satisfies $n_p \equiv 1 \pmod{p}$ and $n_p \mid m$.

For Phase 0, you need fluent use of these statements, not a fresh proof of the Sylow theorems.

3.3 Rings, ideals and quotients

All rings below are commutative with identity unless stated otherwise.

Definition 3.4. An ideal $I \subseteq R$ is an additive subgroup such that $rI \subseteq I$ for every $r \in R$. The quotient R/I consists of cosets $r + I$, with operations inherited from R .

Definition 3.5. A proper ideal $\mathfrak{p}$ is *prime* if $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. A proper ideal $\mathfrak{m}$ is *maximal* if there is no ideal strictly between $\mathfrak{m}$ and R .

The quotient characterizations are fundamental:

$$\mathfrak{p} \text{ prime} \iff R/\mathfrak{p} \text{ domain}, \quad \mathfrak{m} \text{ maximal} \iff R/\mathfrak{m} \text{ field}.$$

Therefore maximal ideals are prime.

3.4 Chinese remainder theorem

Ideals I, J are *comaximal* if $I + J = R$. If $I_1, \dots, I_r$ are pairwise comaximal, then

$$R/\bigcap_{i=1}^r I_i \cong \prod_{i=1}^r R/I_i,$$

and for pairwise comaximal ideals, $\bigcap I_i = \prod I_i$.

Intuition

The Chinese remainder theorem says that mutually independent local congruence conditions can be solved simultaneously and uniquely modulo their combined modulus. Later, local information at different primes will repeatedly be assembled into global information; CRT is the elementary prototype of that philosophy.

A1. Normal subgroups of index two

Prove that every subgroup of index 2 is normal.

A2. Conjugacy in S_3

Compute the centralizer of (12) in S_3 . Use orbit-stabilizer for the conjugation action to recover the size of the conjugacy class of (12) .

A3. Ideals of $\mathbb{Z}$

Prove that every ideal of $\mathbb{Z}$ has the form $n\mathbb{Z}$ for a unique integer $n \geq 0$. Determine when $n\mathbb{Z}$ is prime and when it is maximal.

A4. A quotient-ring computation

Describe the ring $\mathbb{Z}[x]/(2, x^2 + x + 1)$. Is it a field? How many elements does it have?

Chapter 4

Algebra reboot II: polynomials, fields and Galois theory

4.1 Field extensions and degrees

If $K \subseteq L$ are fields, then L is a vector space over K . Its dimension is the extension degree $[L : K]$. For a tower $K \subseteq E \subseteq L$ of finite extensions,

$$[L : K] = [L : E][E : K].$$

If α is algebraic over K , its minimal polynomial $m_{\alpha,K}(x)$ is the unique monic irreducible polynomial in $K[x]$ with $m_{\alpha,K}(\alpha) = 0$. Then

$$K(\alpha) \cong K[x]/(m_{\alpha,K}), \quad [K(\alpha) : K] = \deg m_{\alpha,K}.$$

4.2 Splitting fields, separability and normality

A splitting field of $f \in K[x]$ is the smallest extension in which f factors into linear factors. A finite extension L/K is Galois precisely when it is both normal and separable. In characteristic zero, separability is automatic for algebraic extensions, so for number fields normality is the main extra condition.

Definition 4.1. For a Galois extension L/K ,

$$\text{Gal}(L/K) = \{\sigma : L \rightarrow L : \sigma \text{ is a field automorphism and } \sigma|_K = \text{id}\}.$$

An automorphism is often determined by its action on generators. If $L = K(\alpha)$, then $\sigma(\alpha)$ must be another root of the minimal polynomial of α .

4.3 Finite fields and Frobenius

For a finite field $\mathbb{F}_q$, every element satisfies $x^q = x$. If $L = \mathbb{F}_{q^n}$, then

$$F_q(x) = x^q$$

is an automorphism of L fixing $\mathbb{F}_q$, and

$$\text{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) = \langle F_q \rangle \cong C_n.$$

Intuition

Frobenius is not an arbitrary trick. In a finite field, raising to the q -th power is the canonical symmetry that fixes the base field. Later, for a prime ideal $\mathfrak{p}$, the global Frobenius element will be characterized by reducing modulo a prime above $\mathfrak{p}$ and acting exactly like this finite-field map.

4.4 Fundamental theorem of Galois theory

If L/K is finite Galois with group G , there is an inclusion-reversing correspondence

$$\{K \subseteq E \subseteq L\} \longleftrightarrow \{H \leq G\}$$

given by

$$E \mapsto \text{Gal}(L/E), \quad H \mapsto L^H = \{x \in L : \sigma(x) = x \ \forall \sigma \in H\}.$$

Moreover,

$$[L : E] = |H|, \quad [E : K] = [G : H].$$

The intermediate extension E/K is Galois if and only if $H \trianglelefteq G$, and then

$$\text{Gal}(E/K) \cong G/H.$$

G1. A biquadratic extension

Let $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Prove that $[L : \mathbb{Q}] = 4$, determine $\text{Gal}(L/\mathbb{Q})$, and list all intermediate quadratic fields.

G2. Cyclotomic warm-up

Let $\zeta_5 = e^{2\pi i/5}$. Show that every $\mathbb{Q}$ -automorphism of $\mathbb{Q}(\zeta_5)$ is determined by

$$\zeta_5 \mapsto \zeta_5^a, \quad a \in (\mathbb{Z}/5\mathbb{Z})^\times,$$

and deduce

$$\text{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q}) \cong (\mathbb{Z}/5\mathbb{Z})^\times.$$

G3. Finite-field automorphisms

Determine the fixed field of the automorphism $x \mapsto x^4$ of $\mathbb{F}_{64}$. State the order of this automorphism.

G4. Splitting fields of cubics

Let $f(x) = x^3 - 2$. Explain why its splitting field is $L = \mathbb{Q}(\sqrt[3]{2}, \zeta_3)$, prove $[L : \mathbb{Q}] = 6$, and identify $\text{Gal}(L/\mathbb{Q}) \cong S_3$ by describing the action on the three roots.

Chapter 5

Analysis reboot I: sequences, series and uniform convergence

5.1 Convergence of sequences and series

A sequence $(x_n) \subset \mathbb{C}$ converges to L if for every $\varepsilon > 0$ there exists N such that $|x_n - L| < \varepsilon$ for $n \geq N$.

For series, remember the hierarchy:

$$\text{absolute convergence} \implies \text{convergence},$$

but not conversely.

Useful tests to have readily available:

- comparison and limit comparison;
- ratio and root tests;
- integral test for positive decreasing terms;
- alternating-series test;
- Cauchy criterion.

5.2 Pointwise and uniform convergence

Pointwise convergence allows the threshold N to depend on x :

$$\forall x \forall \varepsilon > 0 \exists N = N(x, \varepsilon) \dots$$

Uniform convergence demands one N work for every x :

$$\forall \varepsilon > 0 \exists N = N(\varepsilon) \forall x \dots$$

Equivalently,

$$f_n \rightarrow f \text{ uniformly} \iff \|f_n - f\|_\infty \rightarrow 0,$$

where the supremum norm is finite on the domain in question.

Theorem 5.1 (Uniform limit theorem). *A uniform limit of continuous functions is continuous.*

Theorem 5.2 (Integration under uniform convergence). *If $f_n \rightarrow f$ uniformly on $[a, b]$ and every f_n is Riemann integrable, then f is integrable and*

$$\int_a^b f_n \rightarrow \int_a^b f.$$

The proof is the simple but powerful estimate

$$\left| \int_a^b (f_n - f) \right| \leq (b - a) \|f_n - f\|_\infty.$$

5.3 Power series

For

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

there exists a radius $R \in [0, \infty]$ such that the series converges absolutely for $|z - z_0| < R$ and diverges for $|z - z_0| > R$. On compact subsets strictly inside the disc of convergence, convergence is uniform.

The Cauchy–Hadamard formula is

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |a_n|^{1/n}.$$

Inside the disc, termwise differentiation and integration are legitimate and preserve the radius of convergence.

R1. Uniform convergence by the M-test

Show that

$$\sum_{n=1}^{\infty} \frac{x^n}{n^2}$$

converges uniformly on $[-1, 1]$. Deduce that its sum is continuous there.

R2. Testing uniform convergence

For $f_n(x) = \frac{x}{1+nx}$ on $[0, 1]$, determine the pointwise limit and decide whether convergence is uniform.

R3. Interchanging sum and integral

Justify rigorously that

$$\int_0^{1/2} \sum_{n=0}^{\infty} x^n dx = \sum_{n=0}^{\infty} \int_0^{1/2} x^n dx,$$

and evaluate both sides.

R4. Radius and boundary

Determine the set of $z \in \mathbb{C}$ for which

$$\sum_{n=1}^{\infty} \frac{z^n}{n}$$

converges absolutely, converges conditionally, or diverges.

Chapter 6

Analysis reboot II: complex analysis

6.1 Holomorphic functions

A function $f : U \rightarrow \mathbb{C}$, with $U \subseteq \mathbb{C}$ open, is holomorphic if the complex derivative exists at every point of U . Complex differentiability is dramatically more rigid than real differentiability: a holomorphic function is automatically analytic, meaning it is locally represented by a convergent power series.

If $f = u + iv$, the Cauchy–Riemann equations are

$$u_x = v_y, \quad u_y = -v_x.$$

Under mild regularity assumptions they characterize holomorphicity.

6.2 Cauchy’s theorem and formula

For a holomorphic function on a suitable simply connected region,

$$\int_{\gamma} f(z) dz = 0$$

for every closed contour γ . The stronger computational form is Cauchy’s integral formula:

$$f(a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - a} dz,$$

when a lies inside γ and f is holomorphic on and inside the contour.

More generally,

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_{\gamma} \frac{f(z)}{(z - a)^{n+1}} dz.$$

Intuition

Cauchy’s formula says that the values of a holomorphic function inside a contour are completely controlled by values on the boundary. This rigidity is why information about a complex analytic function near a line such as $\Re(s) = 1$ can eventually control arithmetic coefficients encoded by a Dirichlet series.

6.3 Zeros, poles and residues

If f has a Laurent expansion near a ,

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n,$$

then the residue is

$$\operatorname{Res}_{z=a} f = c_{-1}.$$

For a simple pole of g/h at a , with $h(a) = 0$ and $h'(a) \neq 0$,

$$\operatorname{Res}_{z=a} \frac{g(z)}{h(z)} = \frac{g(a)}{h'(a)}.$$

Theorem 6.1 (Residue theorem). *If f is meromorphic inside and on a positively oriented simple closed contour γ , with no poles on γ , then*

$$\int_{\gamma} f(z) dz = 2\pi i \sum_{a \text{ inside } \gamma} \operatorname{Res}_{z=a} f.$$

6.4 What to remember for analytic number theory

You will later need confidence with:

- locally uniform convergence of holomorphic functions;
- differentiating convergent power/Dirichlet series in their domain of uniform convergence;
- analytic continuation and uniqueness via the identity theorem;
- meromorphic functions and poles;
- residues and contour estimates;
- logarithmic derivatives such as f'/f .

You do not need to master every contour-integration trick before Phase 1.

C1. Cauchy formula with a derivative

Evaluate

$$\int_{|z|=3} \frac{\sin z}{(z-1)^3} dz.$$

C2. Residue theorem

Evaluate

$$\int_{|z|=2} \frac{z^2}{(z-1)(z+1)(z-3)} dz.$$

C3. Logarithmic derivative

Let f be holomorphic near a , and suppose a is a zero of order m . Prove that f'/f has a simple pole at a with residue m .

C4. Uniform convergence and holomorphicity

Suppose f_n is holomorphic on an open set U , and $f_n \rightarrow f$ uniformly on every compact subset of U . State the theorem that describes f and the convergence of the derivatives. Explain why compact-uniform convergence, rather than merely pointwise convergence, is the natural condition.

Chapter 7

Synthesis: reconnecting the subjects

Phase 0 is not merely a checklist of disconnected skills. The later Chebotarev course repeatedly moves information between algebraic and analytic languages. The following problems force that kind of switching early.

S1. Roots of unity as a finite group

Prove that the set

$$\mu_n = \{z \in \mathbb{C} : z^n = 1\}$$

is a cyclic group under multiplication of order n . Find all generators of μ_{12} .

S2. Polynomial factorization over finite fields

Factor $x^4 - 1$ over $\mathbb{F}_5$ and over $\mathbb{F}_3$. Explain what changes and why repeated factors are related to the derivative.

S3. A first Euler-product identity

For $\Re(s) > 1$, give a rigorous argument that

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p (1 - p^{-s})^{-1}.$$

You may use unique factorization and absolute convergence, but you must state where each is used.

S4. Automorphisms and conjugate roots

Let L/K be a field extension and $\sigma \in \text{Aut}(L/K)$. If $\alpha \in L$ is algebraic over K , prove that $\sigma(\alpha)$ is a root of the minimal polynomial of α over K . Deduce that a K -automorphism of $K(\alpha)$ is determined by the image of α .

S5. Why conjugacy naturally appears

Let a group G act transitively on a set X . For $x, y \in X$, prove that their stabilizers are conjugate subgroups. Explain in one paragraph why this should make you unsurprised that choosing different objects “above” the same base object can produce conjugate, rather than equal, group elements or subgroups.

Chapter 8

A ten-day Phase 0 schedule

This is a suggested structure, not a deadline. If a day's material is already fluent, compress it. If one topic is genuinely missing, spend longer there and shorten a fluent topic elsewhere.

Day	Focus	Work
1	Baseline diagnostic	Complete D1–D18 closed-book. Mark C/U/N before each. Do not grade until finished.
2	Proof reboot	Read Chapter 2; solve P1–P3; reconstruct D3, D6, D17 from memory.
3	Groups/actions	Review Chapter 3 through Sylow; solve A1–A2; redo D4–D5.
4	Rings/ideals	Finish Chapter 3; solve A3–A4; redo D1–D2 and D6.
5	Fields	Read first half of Chapter 4; redo D7; solve G1.
6	Galois/finite fields	Finish Chapter 4; solve G2–G4; redo D8–D9.
7	Real analysis	Read Chapter 5; solve R1–R4; redo D10–D13.
8	Complex analysis	Read Chapter 6; solve C1–C4; redo D14–D15.
9	Synthesis	Solve S1–S5 without solutions. Repair any weak prerequisite discovered.
10	Readiness test	Take Chapter 8 under closed-book conditions, grade using Chapter 9, then make the move-on decision.

Chapter 9

Final readiness test

Allow about 2.5 hours. Use no notes except for the statement of a theorem if a problem explicitly tells you that you may. Each problem is worth 10 points. Rigorous reasoning matters more than speed.

F1. Quotients and maximality

Let $R = \mathbb{Q}[x]$ and $I = (x^2 - 2)$. Prove that I is maximal and identify R/I with a familiar field.

F2. Group action and conjugacy

Let G act transitively on X . Fix $x \in X$. Prove that $|X| = [G : G_x]$ when G is finite. For the conjugation action of S_3 , compute the conjugacy class size of a 3-cycle from its centralizer.

F3. Field degree

Let $\alpha = \sqrt{2} + \sqrt{5}$. Find a polynomial in $\mathbb{Q}[x]$ satisfied by α , prove it is minimal, and determine $[\mathbb{Q}(\alpha) : \mathbb{Q}]$.

F4. Finite-field Frobenius

Let $L = \mathbb{F}_{2^6}$. Determine the order and fixed field of $x \mapsto x^{2^2}$. Justify your answer from the structure of $\text{Gal}(L/\mathbb{F}_2)$.

F5. Galois correspondence

Suppose L/K is Galois with $\text{Gal}(L/K) \cong C_6$. List the possible degrees $[E : K]$ of intermediate fields $K \subseteq E \subseteq L$. Which intermediate extensions E/K are Galois?

F6. Uniform convergence

For $f_n(x) = \frac{x}{n+x}$ on $[0, \infty)$, determine the pointwise limit and decide whether the convergence is uniform. Prove both assertions.

F7. Power series

Find the radius of convergence of

$$\sum_{n=1}^{\infty} \frac{2^n}{n^3} z^n.$$

Classify convergence at every point on the boundary circle.

F8. Complex integration

Evaluate

$$\int_{|z|=2} \frac{e^z}{z^2(z-1)} dz.$$

Give two valid methods if you can.

F9. Proof reconstruction

Prove the first isomorphism theorem for groups, including well-definedness and injectivity of the induced map.

F10. Integration of algebra and analysis

For real $s > 1$, define

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}.$$

Explain rigorously why unique factorization suggests an Euler product over primes, and identify the analytic issue that must be controlled before one is allowed to rearrange the resulting infinite sums/products. You do not need to prove the most general theorem on infinite products, but your argument must be mathematically coherent.

Chapter 10

Readiness rubric: when to move on

Grade the final test out of 100 using the solutions in Part III. Then apply the qualitative conditions below.

Score bands

- **85–100: move on.** Any misses are ordinary rust. Begin Phase 1 and repair isolated gaps when they become relevant.
- **70–84: usually move on.** Inspect the pattern. If weaknesses are localized to one topic, spend 2–4 targeted days there, then begin Phase 1. If failures are spread across every area, repeat selected Phase 0 work first.
- **55–69: short repair cycle.** Spend roughly one additional week on the weakest two areas and retake F1–F10 using altered examples of your own construction.
- **Below 55: rebuild the foundations more deliberately.** This does not mean starting an undergraduate degree again. It means Phase 1 would currently spend too much cognitive effort on prerequisites rather than commutative algebra.

Non-negotiable qualitative gates

Regardless of score, you should be able to do all of the following before moving on:

1. Read a page of proof-heavy graduate mathematics without notation itself being the main obstacle.
2. Use quotient groups/rings and kernels/images without conceptual confusion.
3. Explain prime versus maximal ideals through quotient rings.
4. Compute degrees and minimal polynomials in simple extensions.
5. State the Galois correspondence and use it in finite examples.
6. Explain finite-field Frobenius $x \mapsto x^q$.
7. Distinguish pointwise and uniform convergence and use a supremum estimate.
8. Use Cauchy's integral formula and the residue theorem in standard examples.
9. Reconstruct at least two proofs from a one-line reminder rather than by rereading them.

Checkpoint

Move-on rule. If you score at least 70 and satisfy all nine qualitative gates, begin Phase 1. Do not remain in Phase 0 merely because you cannot solve every difficult exercise smoothly. Later phases will continue to reactivate these skills in context.

Part I

Complete solutions

Chapter 11

Solutions to the baseline diagnostic

Solution to D1

If $n = p$ is prime, every nonzero class $[a] \in \mathbb{Z}/p\mathbb{Z}$ has $p \nmid a$, so $\gcd(a, p) = 1$. Bezout gives integers u, v with $ua + vp = 1$, hence $[u][a] = [1]$. Thus every nonzero class is invertible and $\mathbb{Z}/p\mathbb{Z}$ is a field.

Conversely, if n is composite, write $n = ab$ with $1 < a, b < n$. Then $[a] \neq 0$ and $[b] \neq 0$ in $\mathbb{Z}/n\mathbb{Z}$, but $[a][b] = [n] = 0$. A field has no nonzero zero divisors, so $\mathbb{Z}/n\mathbb{Z}$ is not a field.

Solution to D2

Solve successively. Numbers congruent to 3 (mod 5) and 2 (mod 3) are 8 (mod 15). Write $x = 8 + 15k$. Modulo 7,

$$8 + 15k \equiv 1 + k \equiv 4 \pmod{7},$$

so $k \equiv 3 \pmod{7}$. Thus $x \equiv 8 + 45 = 53 \pmod{105}$.

Ring-theoretically: if $I_1, \dots, I_r$ are pairwise comaximal ideals in a commutative ring R , then the canonical map

$$R \rightarrow \prod_i R/I_i, \quad r \mapsto (r + I_i)_i,$$

is surjective with kernel $\bigcap_i I_i = \prod_i I_i$. Hence

$$R / \prod_i I_i \cong \prod_i R / I_i.$$

Solution to D3

For $k \in \ker \varphi$ and $g \in G$,

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g)^{-1} = e,$$

so $gkg^{-1} \in \ker \varphi$. Thus the kernel is normal.

Define

$$\bar{\varphi} : G / \ker \varphi \rightarrow \operatorname{im} \varphi, \quad g \ker \varphi \mapsto \varphi(g).$$

If $g \ker \varphi = h \ker \varphi$, then $h^{-1}g \in \ker \varphi$, so $\varphi(g) = \varphi(h)$; hence the map is well-defined. It is a homomorphism and is surjective by definition of the image. If $\bar{\varphi}(g \ker \varphi) = e$, then $g \in \ker \varphi$, so the coset is the identity coset. Thus it is injective.

Solution to D4

The natural action is transitive, so the orbit of 1 is $\{1, 2, 3\}$. Its stabilizer consists of permutations fixing 1:

$$(S_3)_1 = \{e, (23)\},$$

which has size 2. Thus $|S_3| = 6 = 3 \cdot 2 = |S_3 \cdot 1| |(S_3)_1|$.

Conjugacy classes in S_3 are determined by cycle type:

$$\{e\}, \quad \{(12), (13), (23)\}, \quad \{(123), (132)\}.$$

Solution to D5

By Sylow, the number n_7 of Sylow 7-subgroups satisfies

$$n_7 \equiv 1 \pmod{7}, \quad n_7 \mid 3.$$

The only possibility is $n_7 = 1$, hence the Sylow 7-subgroup is normal.

Similarly,

$$n_3 \equiv 1 \pmod{3}, \quad n_3 \mid 7,$$

so $n_3 = 1$ or 7. Both are compatible with Sylow's numerical constraints.

Solution to D6

(a) Suppose I is prime. If $(a + I)(b + I) = 0 + I$, then $ab \in I$, hence $a \in I$ or $b \in I$. Thus $a + I = 0$ or $b + I = 0$, so R/I has no nonzero zero divisors. Conversely, if R/I is a domain and $ab \in I$, then $(a + I)(b + I) = 0$, so one factor is zero, meaning $a \in I$ or $b \in I$.
 (b) Suppose I is maximal and $a + I \neq 0$, so $a \notin I$. Then the ideal $I + (a)$ properly contains I , hence equals R . Therefore $1 = i + ra$ for some $i \in I$, $r \in R$, giving $(r + I)(a + I) = 1 + I$. Thus every nonzero class is invertible. Conversely, if R/I is a field and $I \subseteq J \subseteq R$, then J/I is an ideal of the field R/I , hence either 0 or all of R/I . Thus $J = I$ or $J = R$.

Solution to D7

Let $\alpha = \sqrt{2} + \sqrt{3}$. Then

$$\alpha^2 = 5 + 2\sqrt{6},$$

so

$$(\alpha^2 - 5)^2 = 24.$$

Hence α satisfies

$$x^4 - 10x^2 + 1 = 0.$$

To show the polynomial is minimal, observe that $\sqrt{2}, \sqrt{3} \in \mathbb{Q}(\alpha)$: since

$$(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2}) = 1,$$

we have $\alpha^{-1} = \sqrt{3} - \sqrt{2}$, and therefore

$$\sqrt{3} = \frac{\alpha + \alpha^{-1}}{2}, \quad \sqrt{2} = \frac{\alpha - \alpha^{-1}}{2}.$$

Thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Since $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$, the tower has degree $2 \cdot 2 = 4$. Therefore the displayed degree-4 polynomial is minimal and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$.

Solution to D8

In characteristic p , $(x + y)^q = x^q + y^q$ for $q = p^r$, and $(xy)^q = x^q y^q$, so F is a field homomorphism. It is injective because its kernel is zero; on the finite set $\mathbb{F}_{q^n}$, injective implies surjective. For $a \in \mathbb{F}_q$, $a^q = a$, so F fixes the base field. Finally,

$$F^n(x) = x^{q^n} = x$$

for every $x \in \mathbb{F}_{q^n}$, hence $F^n = \text{id}$.

Solution to D9

The roots of $x^3 - 2$ are

$$\sqrt[3]{2}, \quad \zeta_3 \sqrt[3]{2}, \quad \zeta_3^2 \sqrt[3]{2}.$$

Thus $L = \mathbb{Q}(\sqrt[3]{2}, \zeta_3)$. The polynomial is irreducible over $\mathbb{Q}$ by Eisenstein at 2, so $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. This field lies in $\mathbb{R}$, while $\zeta_3 \notin \mathbb{R}$, so adjoining ζ_3 has degree 2. Thus $[L : \mathbb{Q}] = 6$. The Galois group acts faithfully on three roots, giving a subgroup of S_3 of order 6, hence is all of S_3 .

Complex conjugation has order 2 and fixes $\sqrt[3]{2}$. Its fixed field therefore has degree $6/2 = 3$ over $\mathbb{Q}$ and contains $\mathbb{Q}(\sqrt[3]{2})$, which also has degree 3. Hence the fixed field is $\mathbb{Q}(\sqrt[3]{2})$.

Solution to D10

The alternating harmonic series converges by the alternating-series test, but $\sum 1/n$ diverges, so convergence is conditional.

For $\sum_{n \geq 2} 1/[n(\log n)^2]$, the integral test applies:

$$\int_2^\infty \frac{dx}{x(\log x)^2} = \left[-\frac{1}{\log x} \right]_2^\infty = \frac{1}{\log 2} < \infty.$$

Thus it converges absolutely.

For $a_n = n!/n^n$, the ratio test gives

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!}{(n+1)^{n+1}} \frac{n^n}{n!} = \left(\frac{n}{n+1} \right)^n \rightarrow e^{-1} < 1,$$

so the series converges absolutely.

Solution to D11

For $0 \leq x < 1$, $x^n \rightarrow 0$, while $1^n = 1$. Thus

$$f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

The convergence is not uniform. Each f_n is continuous, while f is discontinuous, and a uniform limit of continuous functions is continuous. Alternatively, for any n , choose $x_n = 2^{-1/n} \in (0, 1)$. Then $f(x_n) = 0$ but $f_n(x_n) = 1/2$, so $\|f_n - f\|_\infty \geq 1/2$.

Solution to D12

Uniform convergence means $\|f_n - f\|_\infty \rightarrow 0$. Hence

$$\left| \int_a^b f_n - \int_a^b f \right| \leq \int_a^b |f_n - f| \leq (b-a) \|f_n - f\|_\infty \rightarrow 0.$$

The uniform limit of continuous functions is continuous, so f is integrable on $[a, b]$.

Solution to D13

For coefficients $a_n = n^2/3^n$,

$$\limsup |a_n|^{1/n} = \frac{1}{3} \lim n^{2/n} = \frac{1}{3},$$

so $R = 3$. On $|z| = 3$, the term has magnitude n^2 , hence does not even tend to zero. Therefore the series diverges at every boundary point.

Solution to D14

Both poles 0 and 1 lie inside $|z| = 2$. Using partial fractions,

$$\frac{1}{z(z-1)} = -\frac{1}{z} + \frac{1}{z-1}.$$

Thus Cauchy's formula gives

$$\int_{|z|=2} \frac{e^z}{z(z-1)} dz = 2\pi i(-e^0 + e^1) = 2\pi i(e-1).$$

Solution to D15

Since

$$\frac{e^z}{z^3} = \frac{1 + z + z^2/2 + \dots}{z^3},$$

the coefficient of z^{-1} is $1/2$, so the first residue is $1/2$.

For the second, write

$$\frac{1}{(z^2 + 1)^2} = \frac{1}{(z-i)^2(z+i)^2}.$$

At the double pole $z = i$,

$$\operatorname{Res}_{z=i} f = \frac{d}{dz} \frac{1}{(z+i)^2} \Big|_{z=i} = -\frac{2}{(z+i)^3} \Big|_{z=i} = -\frac{2}{(2i)^3} = \frac{1}{4i} = -\frac{i}{4}.$$

Solution to D16

Assume there are only finitely many primes $p_1, \dots, p_r$. Let

$$N = p_1 p_2 \cdots p_r + 1.$$

Then $N > 1$, so it has a prime divisor q . By the assumed list, $q = p_j$ for some j . But then $q \mid p_1 \cdots p_r$ and $q \mid N$, hence $q \mid 1$, impossible.

This proof does not require the full uniqueness part of unique factorization. It uses existence of a prime divisor for every integer $N > 1$, which can be proved by well-ordering/induction without uniqueness of factorization.

Solution to D17

Proceed by induction on d . A nonzero constant polynomial has no roots. Suppose $d \geq 1$. If f has no roots, we are done. Otherwise choose a root a . By the factor theorem,

$$f(x) = (x - a)g(x)$$

with $\deg g = d - 1$. Every root $b \neq a$ of f satisfies $g(b) = 0$. By induction, g has at most $d - 1$ roots, so f has at most d roots.

Solution to D18

Bolzano–Weierstrass states: every bounded sequence in $\mathbb{R}$ has a convergent subsequence. It does not assert convergence of the whole sequence. For example, $x_n = (-1)^n$ is bounded but does not converge; its even subsequence converges to 1 and its odd subsequence to -1 . Thus subsequential convergence is strictly weaker than convergence of the original sequence.

Chapter 12

Solutions to proof-language exercises

Solution to P1

Uniform convergence is

$$\forall \varepsilon > 0 \exists N \forall n \geq N \forall x \in E : |f_n(x) - f(x)| < \varepsilon.$$

Its negation is

$$\exists \varepsilon_0 > 0 \forall N \exists n \geq N \exists x \in E : |f_n(x) - f(x)| \geq \varepsilon_0.$$

For x^n on $[0, 1]$, the pointwise limit is 0 on $[0, 1)$ and 1 at 1. Set $\varepsilon_0 = 1/2$. For every N , choose $n = N$ and $x = 2^{-1/n} < 1$. Then $f(x) = 0$ while $f_n(x) = 1/2$. This realizes the negation.

Solution to P2

If $y \in \varphi(A \cap B)$, then $y = \varphi(x)$ for some $x \in A \cap B$. Since $x \in A$ and $x \in B$, we have $y \in \varphi(A) \cap \varphi(B)$.

For strictness, take $G = \mathbb{Z} \times \mathbb{Z}$, $\varphi(a, b) = a + b$ into $\mathbb{Z}$, $A = \mathbb{Z} \times \{0\}$, $B = \{0\} \times \mathbb{Z}$. Then $A \cap B = \{(0, 0)\}$, so $\varphi(A \cap B) = \{0\}$, while $\varphi(A) = \varphi(B) = \mathbb{Z}$, hence their intersection is $\mathbb{Z}$.

Solution to P3

Remove compactness but keep continuity: $f(x) = x$ on $(0, 1)$ is continuous but has no maximum; its supremum is 1, not attained.

Remove continuity but keep compactness: define $f : [0, 1] \rightarrow \mathbb{R}$ by $f(x) = x$ for $x < 1$ and $f(1) = 0$. The domain is compact, but f has no maximum: values approach 1 without attaining it.

Chapter 13

Solutions to algebra reboot exercises

Solution to A1

Let $H \leq G$ have index 2. There are exactly two left cosets: H and $G \setminus H$. For $g \notin H$, the left coset gH is therefore $G \setminus H$. Likewise the right coset Hg is not H , so it is also $G \setminus H$. Thus $gH = Hg$ for $g \notin H$; for $g \in H$, both equal H . Hence $H \trianglelefteq G$.

Solution to A2

An element commutes with (12) exactly when conjugation fixes (12) . Directly,

$$C_{S_3}((12)) = \{e, (12)\}.$$

For conjugation, the stabilizer of (12) is its centralizer. Therefore orbit–stabilizer gives

$$|\text{Cl}((12))| = \frac{|S_3|}{|C_{S_3}((12))|} = \frac{6}{2} = 3,$$

as expected from the three transpositions.

Solution to A3

Let $I \neq 0$ be an ideal of $\mathbb{Z}$. Choose the least positive element $n \in I$. Then $n\mathbb{Z} \subseteq I$. For any $a \in I$, division gives $a = qn + r$ with $0 \leq r < n$. Since $r = a - qn \in I$, minimality forces $r = 0$, so $a \in n\mathbb{Z}$. Hence $I = n\mathbb{Z}$. The zero ideal is $0\mathbb{Z}$. Uniqueness follows from equality of positive generators.

For $n > 0$, $n\mathbb{Z}$ is prime iff $\mathbb{Z}/n\mathbb{Z}$ is a domain, iff n is prime. It is maximal iff $\mathbb{Z}/n\mathbb{Z}$ is a field, again iff n is prime. The zero ideal is prime because $\mathbb{Z}$ is a domain, but it is not maximal because $\mathbb{Z}$ is not a field.

Solution to A4

First quotient by (2):

$$\mathbb{Z}[x]/(2, x^2 + x + 1) \cong \mathbb{F}_2[x]/(x^2 + x + 1).$$

The polynomial $x^2 + x + 1$ has no roots in $\mathbb{F}_2$: at 0 it is 1, and at 1 it is $1 + 1 + 1 = 1$. A quadratic over a field is reducible iff it has a root, so it is irreducible. Therefore the quotient is a field with $2^2 = 4$ elements, isomorphic to $\mathbb{F}_4$.

Chapter 14

Solutions to field and Galois exercises

Solution to G1

Since $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$, the tower

$$\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset \mathbb{Q}(\sqrt{2}, \sqrt{3})$$

has degrees 2 and 2, hence $[L : \mathbb{Q}] = 4$. Each automorphism independently sends $\sqrt{2} \mapsto \pm\sqrt{2}$ and $\sqrt{3} \mapsto \pm\sqrt{3}$, yielding four automorphisms. Thus

$$\text{Gal}(L/\mathbb{Q}) \cong C_2 \times C_2.$$

The three index-two subgroups correspond to the three quadratic intermediate fields

$$\mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}(\sqrt{3}), \quad \mathbb{Q}(\sqrt{6}).$$

Solution to G2

The minimal polynomial of ζ_5 is

$$\Phi_5(x) = x^4 + x^3 + x^2 + x + 1,$$

so $[\mathbb{Q}(\zeta_5) : \mathbb{Q}] = 4$. Any automorphism fixes $\mathbb{Q}$ and must send ζ_5 to another root of its minimal polynomial, namely ζ_5^a for $a = 1, 2, 3, 4$. Conversely, each assignment $\zeta_5 \mapsto \zeta_5^a$ with $a \in (\mathbb{Z}/5\mathbb{Z})^\times$ extends to a $\mathbb{Q}$ -automorphism. Composition corresponds to multiplication of exponents mod 5, giving the desired isomorphism.

Solution to G3

We have $64 = 2^6$, and $x \mapsto x^4 = x^{2^2}$ is F^2 , where $F : x \mapsto x^2$ generates the cyclic group $\text{Gal}(\mathbb{F}_{64}/\mathbb{F}_2)$ of order 6. The order of F^2 is

$$\frac{6}{\gcd(6, 2)} = 3.$$

Its fixed field therefore has degree $6/3 = 2$ over $\mathbb{F}_2$, so it is $\mathbb{F}_{2^2} = \mathbb{F}_4$.

Solution to G4

The three roots are $\alpha = \sqrt[3]{2}$, $\zeta_3\alpha$, and $\zeta_3^2\alpha$, so the splitting field is $L = \mathbb{Q}(\alpha, \zeta_3)$. Eisenstein at 2 makes $x^3 - 2$ irreducible, hence $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$. Since $\mathbb{Q}(\alpha) \subset \mathbb{R}$ and $\zeta_3 \notin \mathbb{R}$, adjoining ζ_3 has degree 2, giving $[L : \mathbb{Q}] = 6$.

The Galois group acts faithfully on the three roots, embedding in S_3 . Since it also has order 6, the image is all of S_3 . Concretely, one automorphism cycles the roots by $\alpha \mapsto \zeta_3\alpha$, and complex conjugation fixes α while swapping the other two roots. A 3-cycle and a transposition generate S_3 .

Chapter 15

Solutions to real-analysis exercises

Solution to R1

For $|x| \leq 1$,

$$\left| \frac{x^n}{n^2} \right| \leq \frac{1}{n^2}.$$

Since $\sum 1/n^2$ converges, the Weierstrass M-test gives uniform (indeed absolute uniform) convergence on $[-1, 1]$. Each partial sum is continuous, and a uniform limit of continuous functions is continuous.

Solution to R2

For $x = 0$, $f_n(0) = 0$. For fixed $x > 0$,

$$\frac{x}{1 + nx} \rightarrow 0,$$

so the pointwise limit is zero everywhere. For uniform convergence, note that for $x \geq 0$,

$$0 \leq \frac{x}{1 + nx} < \frac{1}{n}.$$

Indeed $nx < 1 + nx$. Thus

$$\|f_n\|_\infty = \sup_{x \in [0,1]} \frac{x}{1 + nx} = \frac{1}{n+1} \rightarrow 0,$$

where monotonicity in x gives the supremum at $x = 1$. Hence convergence is uniform. This problem is a useful warning that a parameter nx near $x = 0$ does not automatically destroy uniform convergence.

Solution to R3

On $[0, 1/2]$, $|x^n| \leq (1/2)^n$ and $\sum (1/2)^n$ converges. By the M-test, $\sum x^n$ converges uniformly, so termwise integration is valid. The sum is $1/(1-x)$, hence

$$\int_0^{1/2} \frac{dx}{1-x} = \log 2.$$

On the other hand,

$$\sum_{n=0}^{\infty} \int_0^{1/2} x^n dx = \sum_{n=0}^{\infty} \frac{2^{-(n+1)}}{n+1} = \sum_{m=1}^{\infty} \frac{(1/2)^m}{m} = \log 2.$$

Solution to R4

The radius is 1, since $|1/n|^{1/n} \rightarrow 1$. If $|z| < 1$, the series converges absolutely. If $|z| > 1$, the terms z^n/n do not tend to zero.

On $|z| = 1$, absolute convergence would require $\sum 1/n$, so it never converges absolutely. At $z = 1$, it is the harmonic series and diverges. At every other point $z = e^{i\theta} \neq 1$, the partial sums of z^n are bounded:

$$\sum_{n=1}^N z^n = z \frac{1 - z^N}{1 - z},$$

and $1/n \searrow 0$. Dirichlet's test therefore gives convergence. Thus it converges conditionally at all boundary points except $z = 1$.

Chapter 16

Solutions to complex-analysis exercises

Solution to C1

Cauchy's differentiation formula gives

$$\int_{|z|=3} \frac{\sin z}{(z-1)^3} dz = \frac{2\pi i}{2!} (\sin z)''|_{z=1} = \pi i (-\sin 1).$$

Thus the value is $-\pi i \sin 1$.

Solution to C2

The poles inside $|z| = 2$ are 1 and -1 ; the pole 3 lies outside. The residues are

$$\operatorname{Res}_{z=1} f = \frac{1}{(1+1)(1-3)} = -\frac{1}{4},$$

and

$$\operatorname{Res}_{z=-1} f = \frac{(-1)^2}{(-1-1)(-1-3)} = \frac{1}{8}.$$

Their sum is $-1/8$, so the integral is

$$2\pi i \left(-\frac{1}{8}\right) = -\frac{\pi i}{4}.$$

Solution to C3

If a is a zero of order m , then locally

$$f(z) = (z-a)^m g(z),$$

where g is holomorphic and $g(a) \neq 0$. Hence

$$\frac{f'(z)}{f(z)} = \frac{m}{z-a} + \frac{g'(z)}{g(z)}.$$

Because $g(a) \neq 0$, g'/g is holomorphic near a . Thus f'/f has a simple pole at a with residue m .

Solution to C4

Weierstrass' theorem states that a locally uniform (uniform on compact subsets) limit of holomorphic functions is holomorphic. Moreover, for each $k \geq 1$, the derivatives satisfy $f_n^{(k)} \rightarrow f^{(k)}$ uniformly on compact subsets.

Pointwise convergence gives no uniform control on a contour surrounding a point, so one cannot safely pass a limit through Cauchy's integral formula. Compact-uniform convergence supplies exactly that control on every closed contour contained in U , allowing the limit and contour integral to be interchanged.

Chapter 17

Solutions to synthesis problems

Solution to S1

Closure, identity and inverses are immediate: if $z^n = w^n = 1$, then $(zw)^n = 1$, and $(z^{-1})^n = 1$. The roots are

$$\mu_n = \{e^{2\pi i k/n} : 0 \leq k < n\} = \langle e^{2\pi i/n} \rangle,$$

so the group is cyclic of order n . An element ζ_n^k is a generator iff $\gcd(k, n) = 1$. Thus generators of μ_{12} are ζ_{12}^k for $k \in \{1, 5, 7, 11\}$.

Solution to S2

Over $\mathbb{F}_5$,

$$x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x - 2)(x + 2),$$

because $2^2 \equiv -1 \pmod{5}$. Thus it splits into four distinct linear factors.

Over $\mathbb{F}_3$,

$$x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x^2 + 1),$$

and $x^2 + 1$ has no root in $\mathbb{F}_3$, so it is irreducible.

Repeated factors are detected by $\gcd(f, f')$: a root a is repeated precisely when $f(a) = f'(a) = 0$. Here $f'(x) = 4x^3$, which is nonzero at nonzero roots in characteristics 3 and 5, so the factors are squarefree in both cases.

Solution to S3

For real part $\sigma = \Re(s) > 1$, the series

$$\sum_{n \geq 1} |n^{-s}| = \sum n^{-\sigma}$$

converges, so rearrangements of the associated absolutely convergent sums are legitimate. For a finite set of primes P , expand the finite product of geometric series:

$$\prod_{p \in P} (1 - p^{-s})^{-1} = \prod_{p \in P} \sum_{k \geq 0} p^{-ks} = \sum_{n: \text{all prime factors of } n \in P} n^{-s}.$$

Unique factorization is used to ensure each such n appears exactly once, corresponding to its vector of prime exponents. Letting P increase over all primes, the right-hand sums increase in coverage to all positive integers; absolute convergence controls the limiting

process. Hence

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}.$$

For complex s , the same argument is justified by absolute convergence at $\sigma > 1$.

Solution to S4

Let $m(x) = \sum a_j x^j \in K[x]$ be the minimal polynomial of α . Since σ fixes every coefficient $a_j \in K$,

$$m(\sigma(\alpha)) = \sum a_j \sigma(\alpha)^j = \sigma \left(\sum a_j \alpha^j \right) = \sigma(m(\alpha)) = 0.$$

Thus $\sigma(\alpha)$ is another root of m . Every element of $K(\alpha)$ is a rational expression in α with coefficients in K , so once $\sigma(\alpha)$ is known, σ is determined on all of $K(\alpha)$.

Solution to S5

Because the action is transitive, choose $g \in G$ with $gx = y$. Then

$$G_y = gG_xg^{-1}.$$

Indeed, if $h \in G_x$, then

$$(ghg^{-1})y = ghx = gx = y,$$

so $gG_xg^{-1} \subseteq G_y$, and applying the same argument with g^{-1} gives equality.

Conceptually, an object chosen above a base object is often noncanonical: another choice is obtained by a symmetry g . Structures defined relative to the choice are therefore transported by g , which algebraically appears as conjugation. Chebotarev later exhibits exactly this phenomenon when different primes $\mathfrak{P}$ above the same $\mathfrak{p}$ yield conjugate Frobenius elements.

Chapter 18

Solutions and marking guide for the final readiness test

Each problem is worth 10 points. Award approximately 7 points for a correct core argument with minor omissions; 4–6 for substantial correct structure but a serious gap; 1–3 for relevant fragments; 0 for no meaningful approach.

Solution to F1

The polynomial $x^2 - 2$ is irreducible over $\mathbb{Q}$: a reducible quadratic would have a rational root, but $\sqrt{2} \notin \mathbb{Q}$ (for instance by the standard parity proof). Since $\mathbb{Q}[x]$ is a PID and an irreducible polynomial generates a maximal ideal, $(x^2 - 2)$ is maximal. Equivalently,

$$\mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt{2}),$$

via evaluation $f(x) \mapsto f(\sqrt{2})$; the kernel is $(x^2 - 2)$, and the target is a field. **10 points:** 3 irreducibility, 4 quotient-field argument, 3 correct identification/isomorphism.

Solution to F2

The map

$$G/G_x \rightarrow X, \quad gG_x \mapsto gx$$

is well-defined, injective, and surjective by transitivity, so $|X| = [G : G_x]$. For conjugation of S_3 , take $c = (123)$. Its centralizer is $\langle c \rangle = \{e, c, c^2\}$, of size 3. Hence its conjugacy class has size $6/3 = 2$. **10 points:** 6 for the coset bijection proof, 4 for the conjugacy computation.

Solution to F3

Let $\alpha = \sqrt{2} + \sqrt{5}$. Then

$$\alpha^2 = 7 + 2\sqrt{10},$$

so $(\alpha^2 - 7)^2 = 40$, giving

$$\alpha^4 - 14\alpha^2 + 9 = 0.$$

To prove degree 4, note

$$(\sqrt{5} + \sqrt{2})(\sqrt{5} - \sqrt{2}) = 3,$$

so $3/\alpha = \sqrt{5} - \sqrt{2}$. Hence

$$\sqrt{5} = \frac{1}{2} \left(\alpha + \frac{3}{\alpha} \right), \quad \sqrt{2} = \frac{1}{2} \left(\alpha - \frac{3}{\alpha} \right).$$

Thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{5})$, which has degree 4 because $\sqrt{5} \notin \mathbb{Q}(\sqrt{2})$. Therefore $x^4 - 14x^2 + 9$ is minimal. **10 points:** 4 polynomial, 4 minimality/degree, 2 field identification.

Solution to F4

Let $F : x \mapsto x^2$, a generator of $\text{Gal}(\mathbb{F}_{2^6}/\mathbb{F}_2) \cong C_6$. The given map is F^2 , whose order is

$$6/\gcd(6, 2) = 3.$$

Its fixed field therefore has degree $6/3 = 2$ over $\mathbb{F}_2$, hence is $\mathbb{F}_{2^2} = \mathbb{F}_4$. Equivalently, $x^4 = x$ holds exactly on $\mathbb{F}_4$ inside $\mathbb{F}_{64}$. **10 points:** 5 order, 5 fixed field with justification.

Solution to F5

A cyclic group C_6 has exactly one subgroup of each order dividing 6: 1, 2, 3, 6. If $H = \text{Gal}(L/E)$, then

$$[E : K] = [G : H] = 6/|H|,$$

so possible degrees are 1, 2, 3, 6. Every subgroup of an abelian group is normal, so every intermediate extension E/K is Galois. **10 points:** 6 for degrees/correspondence, 4 for normality/Galois conclusion.

Solution to F6

For each fixed $x \geq 0$,

$$f_n(x) = \frac{x}{n+x} \rightarrow 0.$$

However, convergence is not uniform on the unbounded domain. For each n , as $x \rightarrow \infty$,

$$\frac{x}{n+x} \rightarrow 1,$$

so $\sup_{x \geq 0} f_n(x) = 1$. Hence $\|f_n - 0\|_\infty = 1$ for all n , not tending to zero. **10 points:** 4 pointwise limit, 6 uniform analysis.

Solution to F7

Here $a_n = 2^n/n^3$, so

$$\limsup |a_n|^{1/n} = 2,$$

and $R = 1/2$. On $|z| = 1/2$, write $z = (1/2)e^{i\theta}$. Then

$$\left| \frac{2^n}{n^3} z^n \right| = \frac{1}{n^3}.$$

Thus the series converges absolutely at every boundary point. **10 points:** 5 radius, 5 complete boundary classification.

Solution to F8

The poles inside $|z| = 2$ are 0 (order two) and 1 (simple). Partial fractions are unnecessary. At 1,

$$\operatorname{Res}_{z=1} \frac{e^z}{z^2(z-1)} = e.$$

At 0, write $e^z/(z-1) = g(z)$, so the integrand is $g(z)/z^2$ and the residue is $g'(0)$. Since

$$g'(z) = \frac{e^z(z-1) - e^z}{(z-1)^2} = \frac{e^z(z-2)}{(z-1)^2},$$

we get $g'(0) = -2$. Therefore

$$\int_{|z|=2} \frac{e^z}{z^2(z-1)} dz = 2\pi i(e-2).$$

A second method is to decompose $1/[z^2(z-1)]$ into partial fractions and apply Cauchy's formula. **10 points:** 7 correct residues, 3 final theorem/orientation.

Solution to F9

Define

$$\bar{\varphi} : G/\ker \varphi \rightarrow \operatorname{im} \varphi, \quad g \ker \varphi \mapsto \varphi(g).$$

First, $\ker \varphi \trianglelefteq G$, since $\varphi(gkg^{-1}) = e$ for $k \in \ker \varphi$. If $g \ker \varphi = h \ker \varphi$, then $h^{-1}g \in \ker \varphi$, so $\varphi(g) = \varphi(h)$: well-definedness. The map is a homomorphism because

$$\bar{\varphi}((gK)(hK)) = \varphi(gh) = \varphi(g)\varphi(h).$$

It is surjective by definition of $\operatorname{im} \varphi$. If $\bar{\varphi}(gK) = e$, then $g \in K = \ker \varphi$, so $gK = K$, proving injectivity. **10 points:** 2 normality, 3 well-definedness, 2 homomorphism/surjectivity, 3 injectivity.

Solution to F10

For $s > 1$, each prime contributes a geometric series

$$(1 - p^{-s})^{-1} = 1 + p^{-s} + p^{-2s} + \dots$$

For finitely many primes, multiplying these series gives one term

$$p_1^{-a_1 s} \dots p_r^{-a_r s} = n^{-s}$$

for every integer whose prime factors lie among those primes. Unique factorization ensures a one-to-one correspondence between integers and exponent vectors $(a_p)_p$, so formally the product over all primes should enumerate every n^{-s} exactly once.

The analytic issue is that an infinite product expansion entails rearranging/limiting infinitely many sums. Such manipulations are not automatically valid for conditionally convergent expressions. Here absolute convergence

$$\sum_{n=1}^{\infty} n^{-s} < \infty$$

for $s > 1$ supplies the required control; finite Euler products converge toward the full Dirichlet series, and rearrangements are legitimate in the absolutely convergent region. **10 points:** 4 UFD mechanism, 4 analytic convergence issue, 2 coherent limiting argument.

What comes next

When you satisfy the readiness gate, begin Phase 1: targeted commutative algebra. The conceptual chain you are preparing for is

ideals $\rightarrow$ localization $\rightarrow$ integrality $\rightarrow$ Dedekind domains $\rightarrow$ prime ideal factorization.

Do not wait until every topic in this workbook feels effortless. The next phase will reinforce quotient rings, field extensions, and proof technique in a more meaningful context.

Compact reference sheet

Algebra

$$\begin{aligned}G/\ker \varphi &\cong \operatorname{im} \varphi, \\|G| &= |Gx| |G_x|, \\ \mathfrak{p} \text{ prime} &\iff R/\mathfrak{p} \text{ domain}, \\ \mathfrak{m} \text{ maximal} &\iff R/\mathfrak{m} \text{ field}, \\ [L : K] &= [L : E][E : K], \\ [L : L^H] &= |H| \quad (L/K \text{ finite Galois}), \\ \operatorname{Gal}(\mathbb{F}_{q^n}/\mathbb{F}_q) &= \langle x \mapsto x^q \rangle \cong C_n.\end{aligned}$$

Analysis

$$\begin{aligned}f_n \rightarrow f \text{ uniformly} &\iff \|f_n - f\|_\infty \rightarrow 0, \\ \frac{1}{R} &= \limsup |a_n|^{1/n}, \\ f(a) &= \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z-a} dz, \\ f^{(n)}(a) &= \frac{n!}{2\pi i} \int_\gamma \frac{f(z)}{(z-a)^{n+1}} dz, \\ \int_\gamma f(z) dz &= 2\pi i \sum \operatorname{Res} f.\end{aligned}$$

Proof prompts

When stuck, ask:

- What exactly are the quantifiers?
- What would a witness look like?
- Can I unfold the definition?
- Would the contrapositive produce a witness?
- Is there a natural homomorphism? What is its kernel?
- Is there a group action? What are orbit and stabilizer?
- Is the statement about an extension degree? Can I insert an intermediate field?
- Is an interchange of limit/sum/integral being used? What convergence theorem justifies it?